

Routing

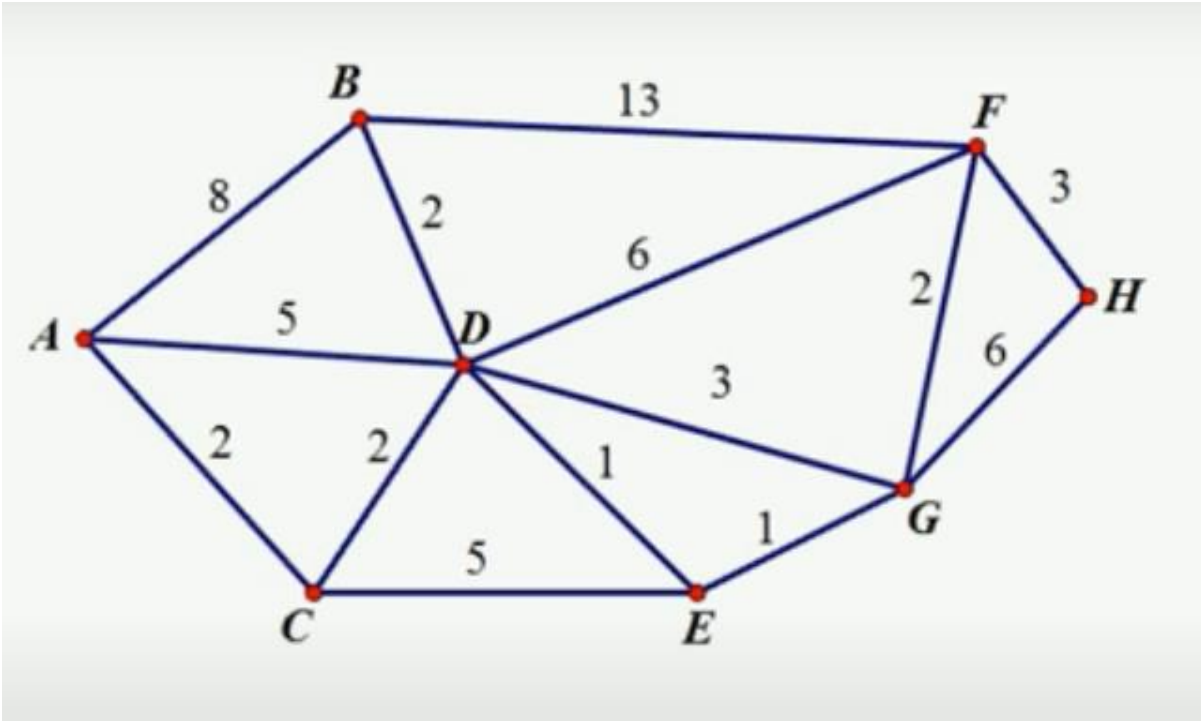
Routing protocols

- Unicast routing protocol
 - Open source shortest path – based on link state
 - Routing information protocol – based on the distance vector

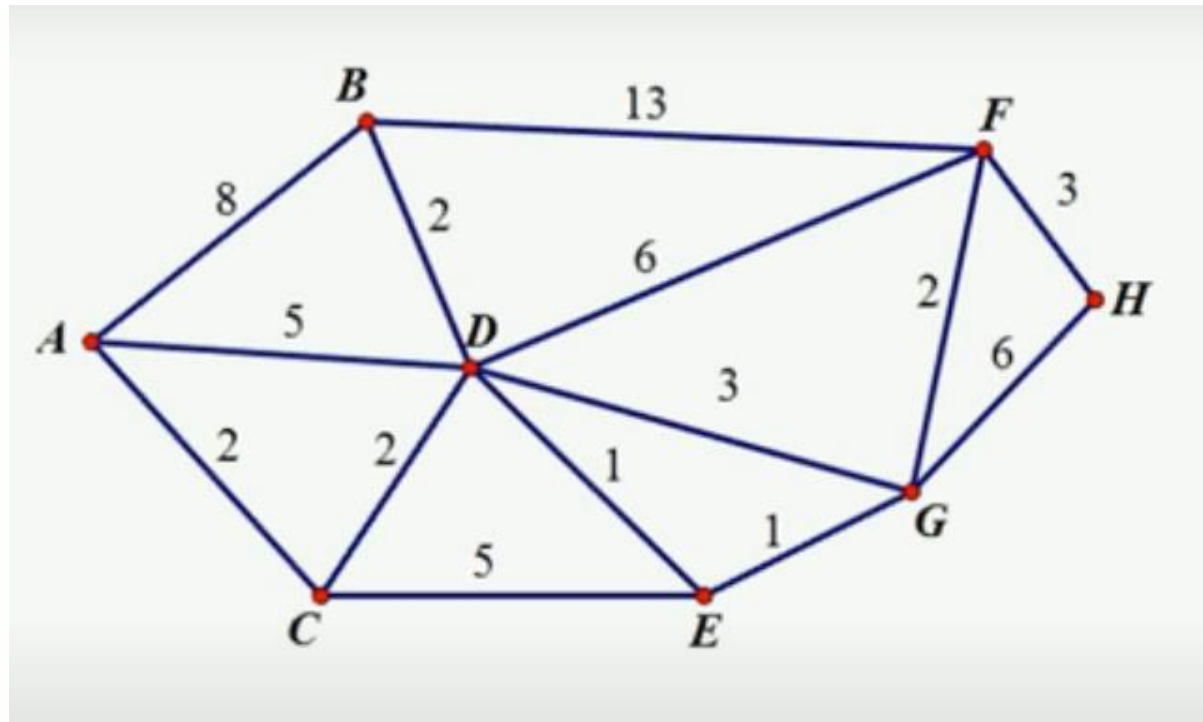
Link state routing

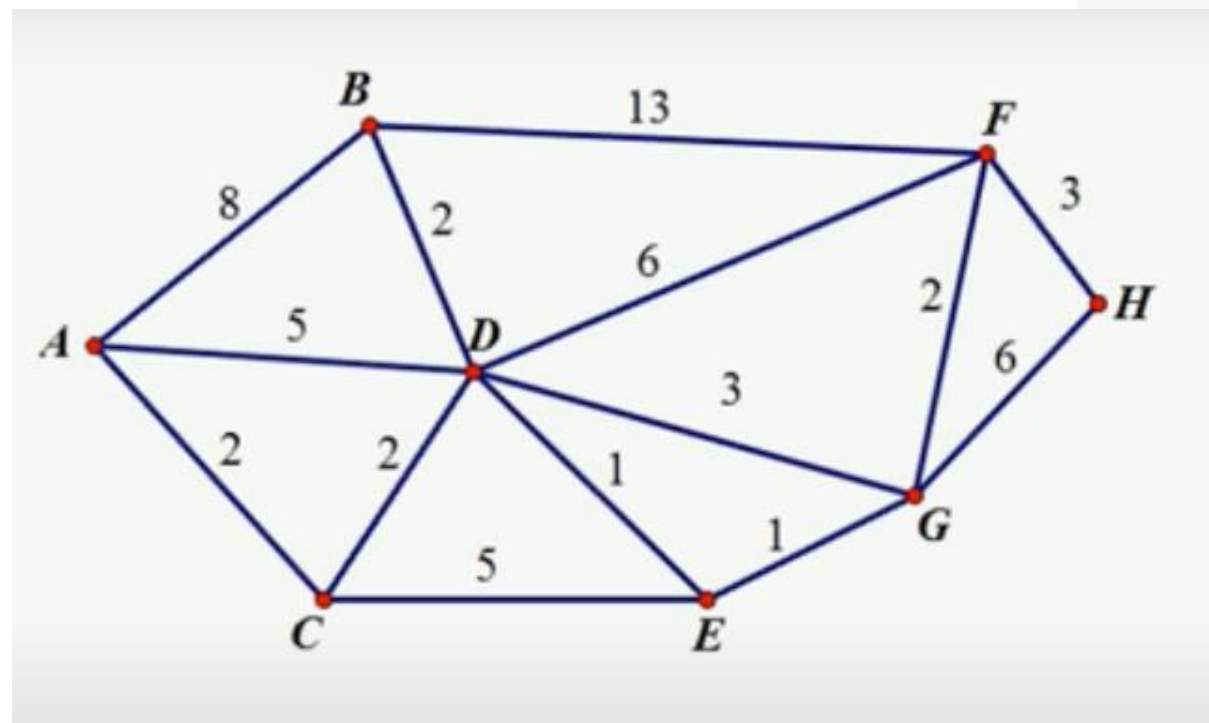
- also known as Dijkstra's Algorithm
- It computes the least cost path from one node (source node) to all other nodes in the network
- it is iterative until k^{th} least cost paths are known to k destination nodes

<i>V</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>
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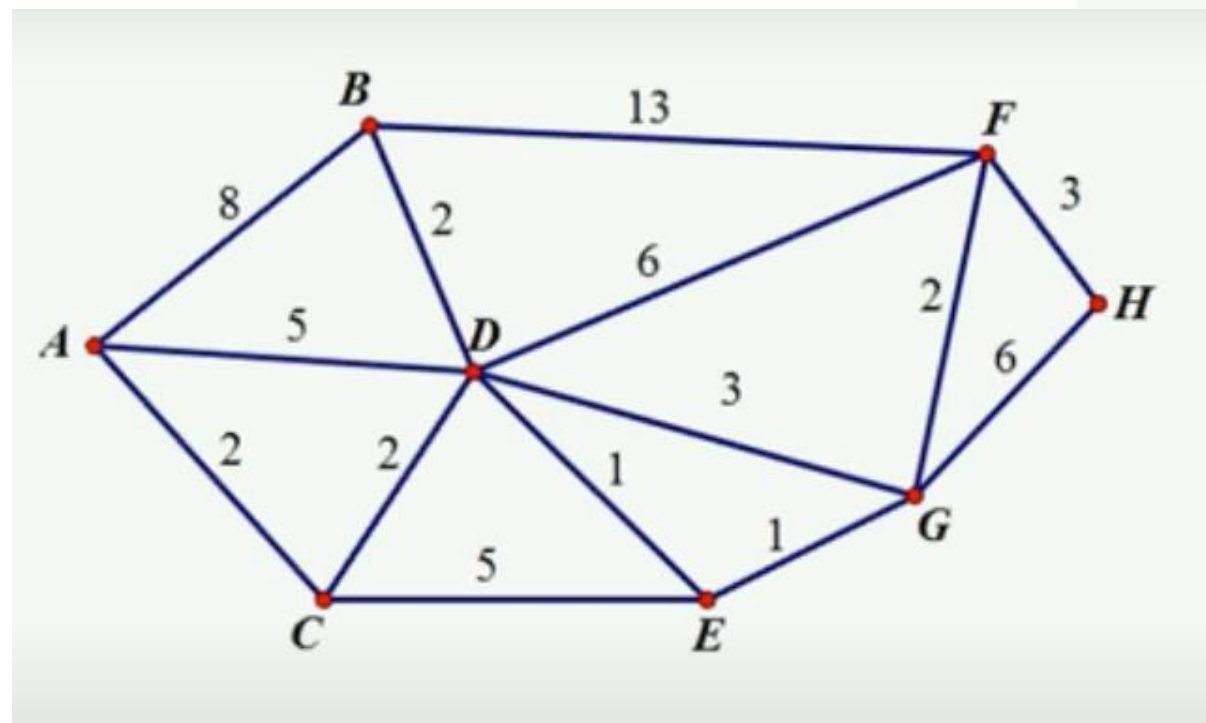


v	A	B	C	D	E	F	G	H
A	0_A							



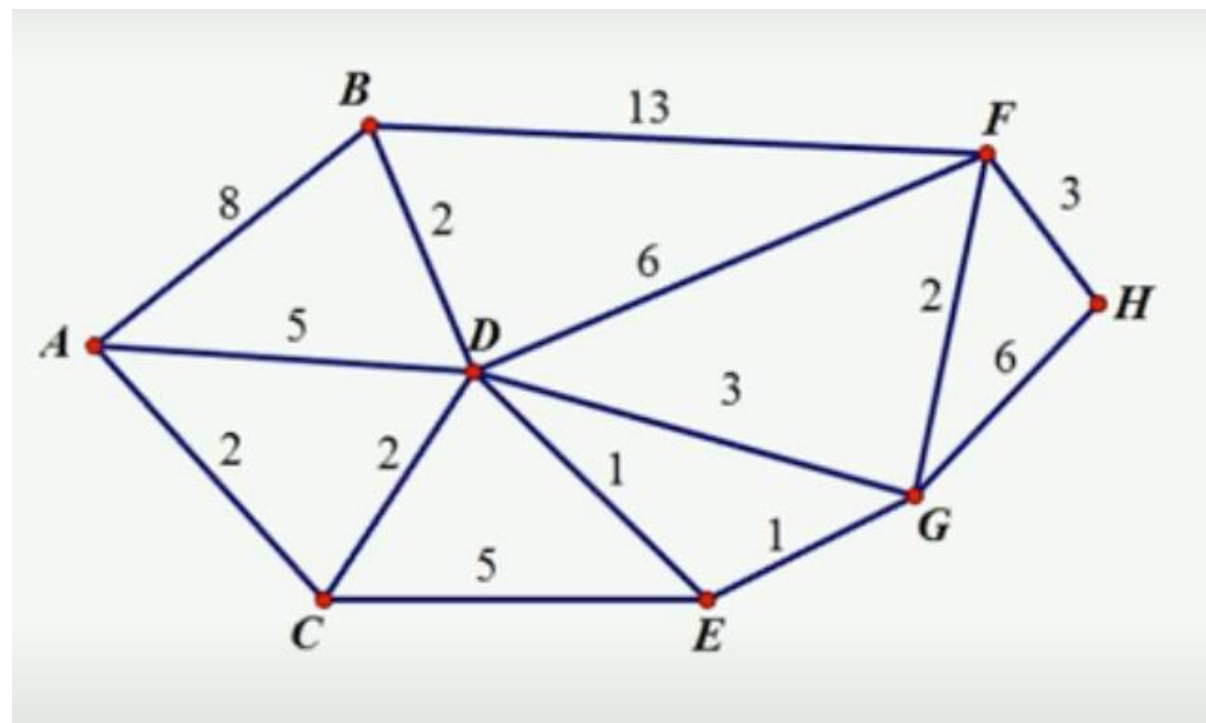


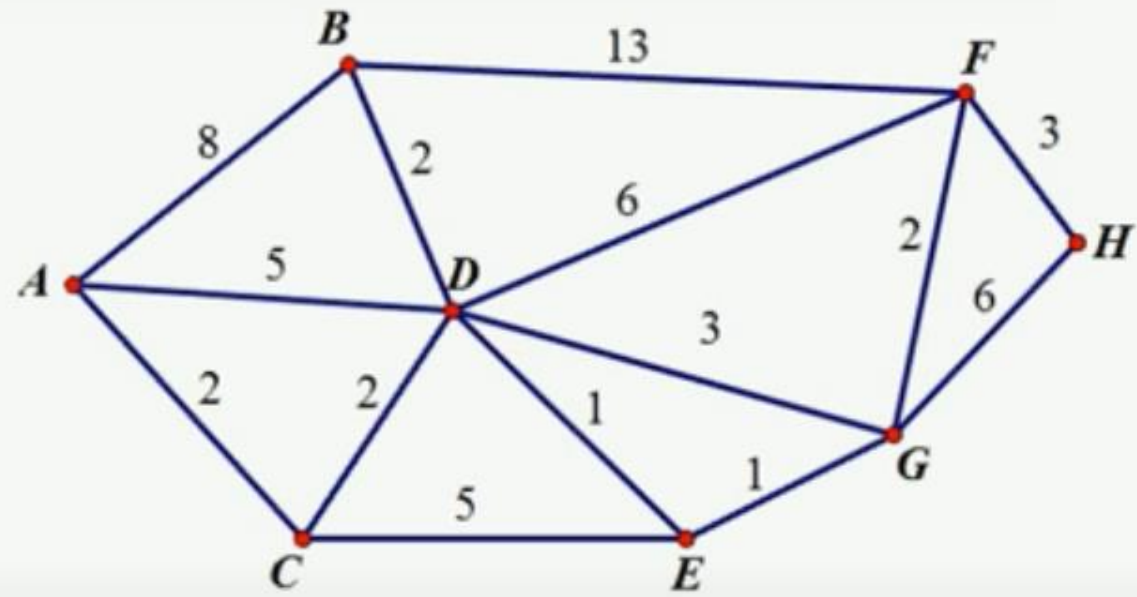
v	A	B	C	D	E	F	G	H
A	0_A	8_A	2_A	5_A	∞	∞	∞	∞
			2_A					



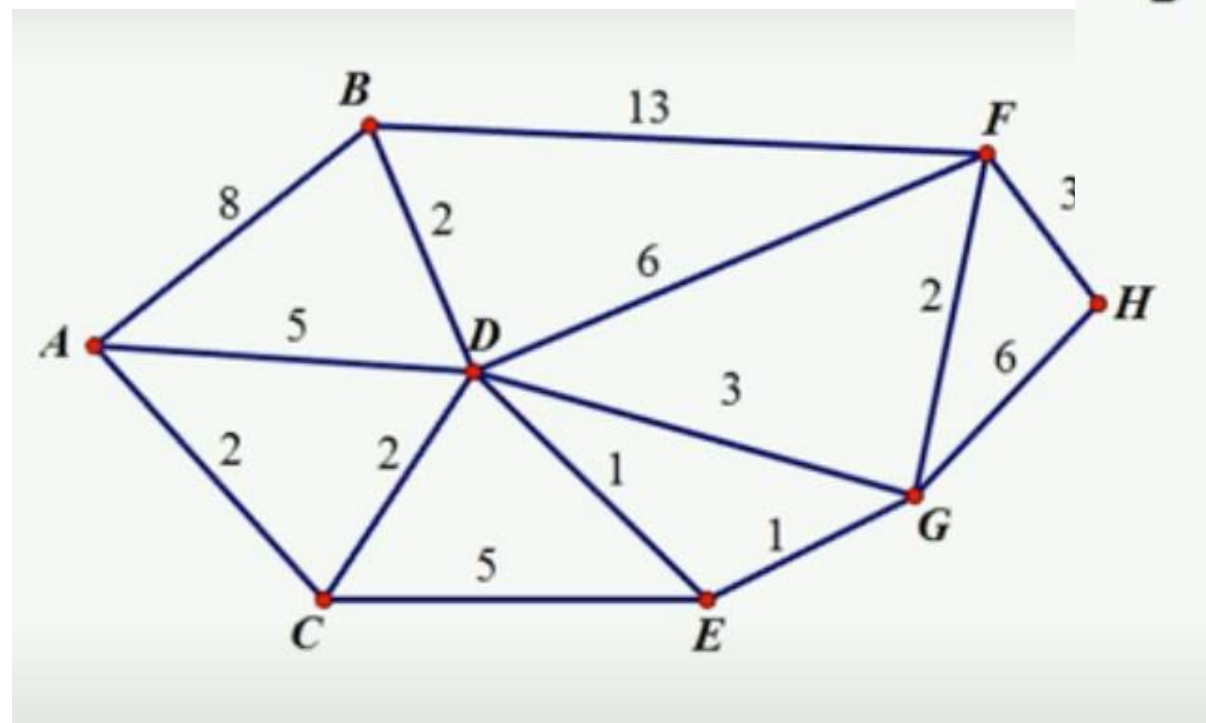
<i>v</i>	A	B	C	D	E	F	G	H
A	0_A	8_A	2_A	5_A	∞	∞	∞	∞
C			2_A					

v	A	B	C	D	E	F	G	H
A	0_A	8_A	2_A	5_A	∞	∞	∞	∞
C		8_A	2_A	4_C	7_C	∞	∞	∞

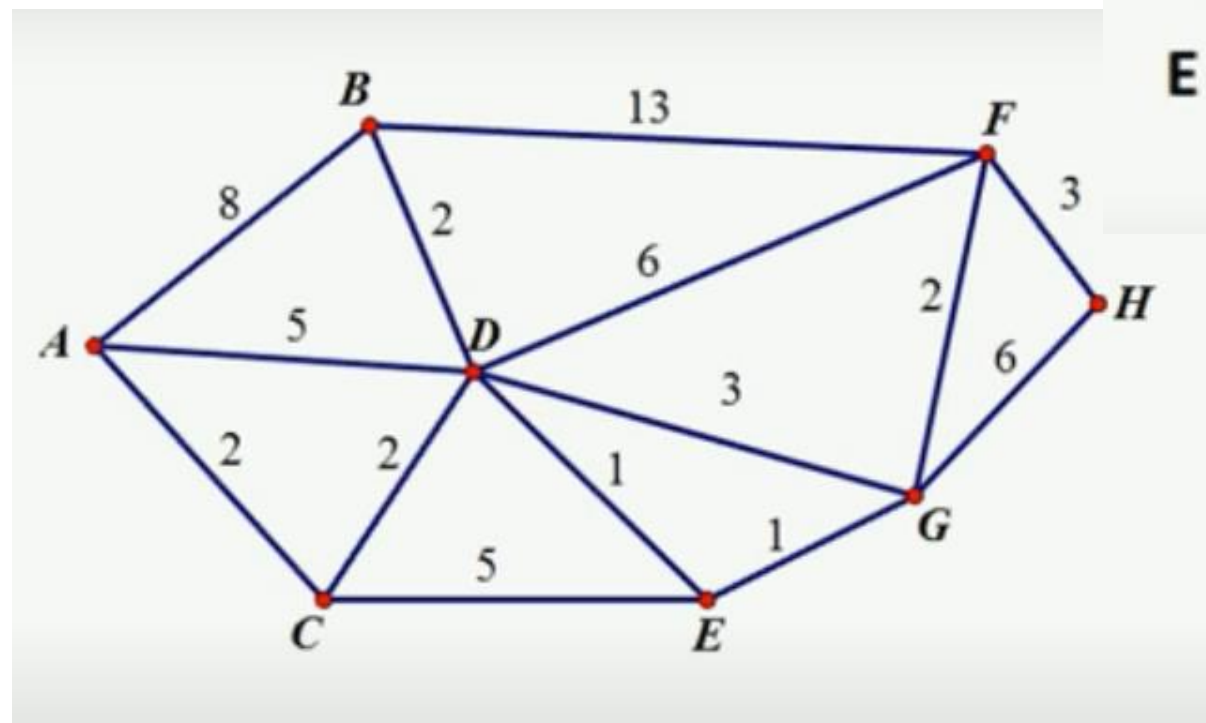




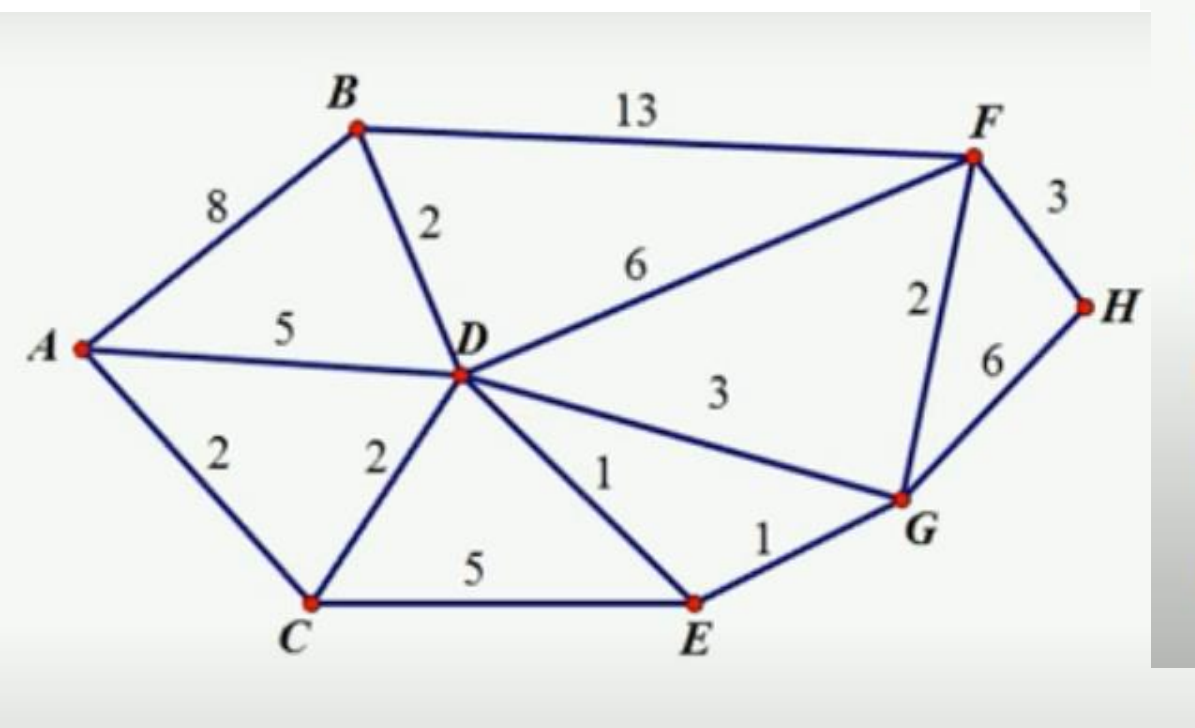
<i>v</i>	A	B	C	D	E	F	G	H
A	0_A	8_A	2_A	5_A	∞	∞	∞	∞
C		8_A	2_A	4_C	7_C	∞	∞	∞
D				4_C				



<i>v</i>	A	B	C	D	E	F	G	H
A	0_A	8_A	2_A	5_A	∞	∞	∞	∞
C		8_A	2_A	4_C	7_C	∞	∞	∞
D		6_D		4_C	5_D	10_D	7_D	∞

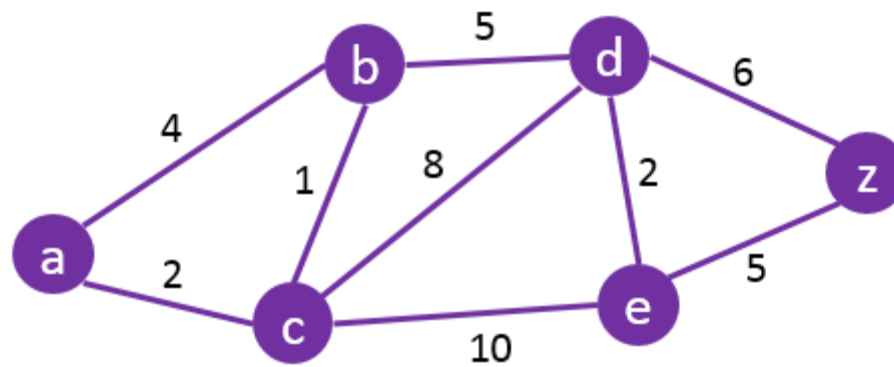


V	A	B	C	D	E	F	G	H
A	0_A	8_A	2_A	5_A	∞	∞	∞	∞
C		8_A	2_A	4_C	7_C	∞	∞	∞
D		6_D		4_C	5_D	10_D	7_D	∞
E		6_D			5_D	10_D	6_E	∞



V	A	B	C	D	E	F	G	H	.
A	0_A	8_A	2_A	5_A	∞	∞	∞	∞	
C		8_A	2_A	4_C	7_C	∞	∞	∞	
D		6_D		4_C	5_D	10_D	7_D	∞	
E		6_D			5_D	10_D	6_E	∞	
B		6_D				10_D	6_E	∞	
G						8_G	6_E	12_G	
F						8_G		11_F	
H								11_F	

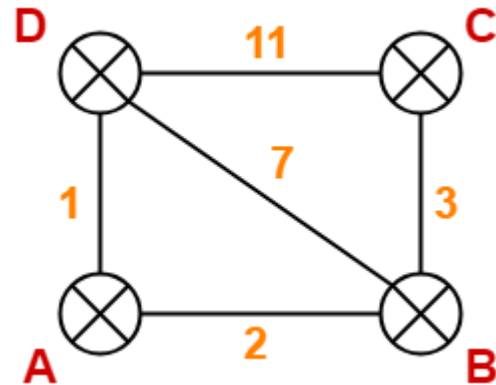
- Apply link state routing and find the shortest path from A to Z?



Distance vector routing

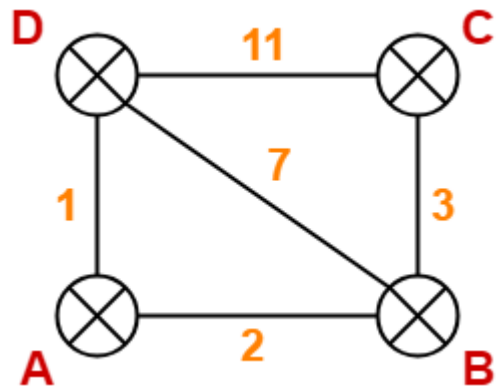
- Distance Vector Routing is a dynamic routing algorithm.
- Follows Bellman ford algorithm
- Each router **prepares its routing table (step 1)**. By their local knowledge. each router knows about,
 - All the routers present in the network
 - Distance to its neighboring routers

Step 1



Router A

Destn	Dist	Nxt hop
A	0	A
B	2	B
C	∞	—
D	1	D



Router D

Destn	Dist	Nxt hop
A	1	A
B	7	B
C	11	C
D	0	D

Router B

Destn	Dist	Nxt hop
A	2	A
B	0	B
C	3	C
D	7	D

Router C

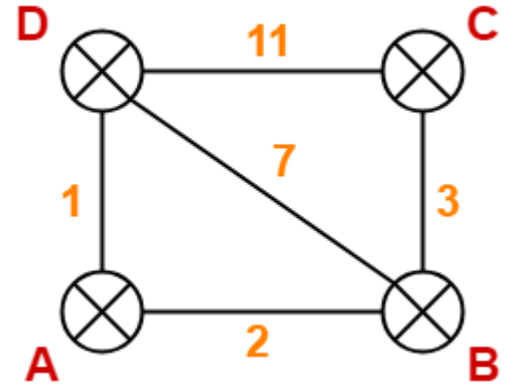
Destn	Dist	Nxt hop
A	∞	—
B	3	B
C	0	C
D	11	D

• Step - 2

- Each router exchanges its distance vector obtained in Step-1 with its neighbors.
- After exchanging the distance vectors, each router prepares a new routing table.

Router A

- Router A receives distance vectors from its **neighbors B and D**.
- Router A prepares a new routing table as



Old table, Router A

Destn	Dist	Nxt hop
A	0	A
B	2	B
C	∞	—
D	1	D

From B

A	2
B	0
C	3
D	7

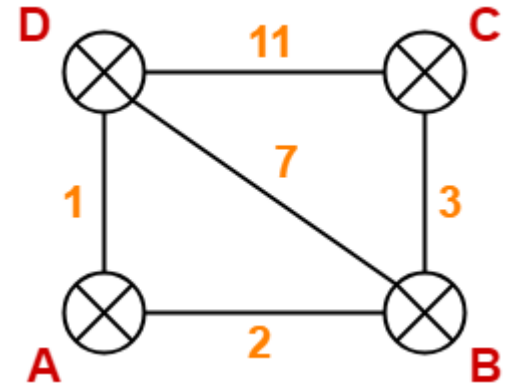
From D

1
7
11
0

New table, Router A

Destn	Dist	Nxt hop
A	0	A
B	2	B
C	∞	--
D	1	D

- Cost of reaching router B from router A **via neighbor B** = Cost (A→B) + Cost (B→B) = **2 + 0**
- Cost of reaching router B from router A **via neighbor D** = Cost (A→D) + Cost (D→B) = **1 + 7**
- $\min \{ 2+0, 1+7 \} = 2$ via B



Old table, Router A

Destn	Dist	Nxt hop
A	0	A
B	2	B
C	∞	—
D	1	D

From B

A	2
B	0
C	3
D	7

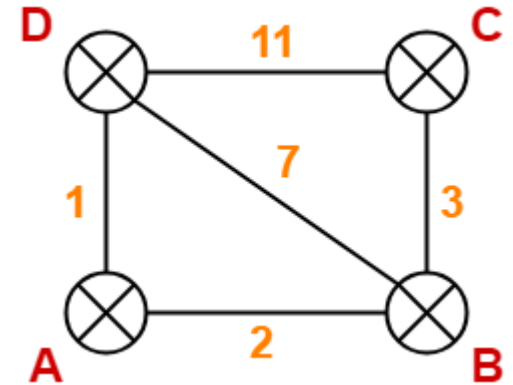
From D

1
7
11
0

New table, Router A

Destn	Dist	Nxt hop
A	0	A

- Cost of reaching router B from router A **via neighbor B** = Cost (A→B) + Cost (B→B) = **2 + 0**
- Cost of reaching router B from router A **via neighbor D** = Cost (A→D) + Cost (D→B) = **1 + 7**
- **min { 2+0 , 1+7 } = 2 via B**



Old table, Router A

Destn	Dist	Nxt hop
A	0	A
B	2	B
C	∞	—
D	1	D

From B

A	2
B	0
C	3
D	7

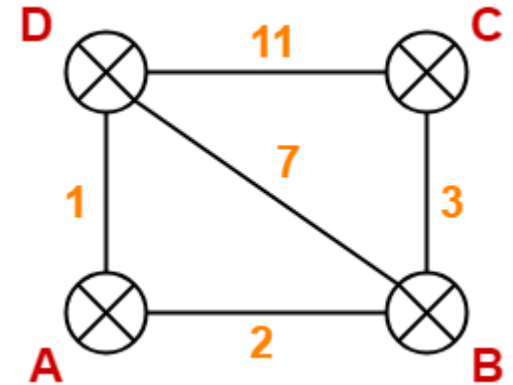
From D

1
7
11
0

New table, Router A

Destn	Dist	Nxt hop
A	0	A
B	2	B

- Cost of reaching router C from router A **via neighbor B** = Cost (A→B) + Cost (B→C) = **2 + 3**
- Cost of reaching router C from router A **via neighbor D** = Cost (A→D) + Cost (D→C) = **1 + 11**
- **min { 2+3 , 1+11 } = 5 via B**



Old table, Router A

Destn	Dist	Nxt hop
A	0	A
B	2	B
C	∞	—
D	1	D

From B

A	2
B	0
C	3
D	7

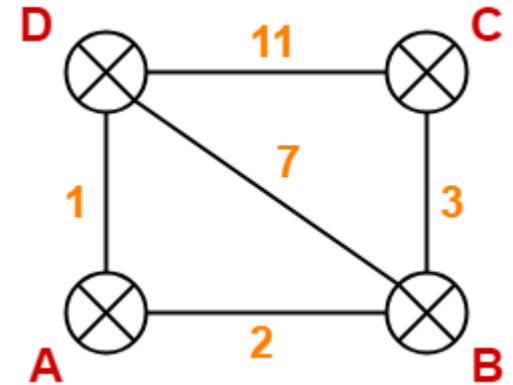
From D

1
7
11
0

New table, Router A

Destn	Dist	Nxt hop
A	0	A
B	2	B

- Cost of reaching router C from router A **via neighbor B** = Cost (A→B) + Cost (B→C) = **2 + 3**
- Cost of reaching router C from router A **via neighbor D** = Cost (A→D) + Cost (D→C) = **1 + 11**
- **min { 2+3 , 1+11 } = 5 via B**



Old table, Router A

Destn	Dist	Nxt hop
A	0	A
B	2	B
C	∞	—
D	1	D

From B

A	2
B	0
C	3
D	7

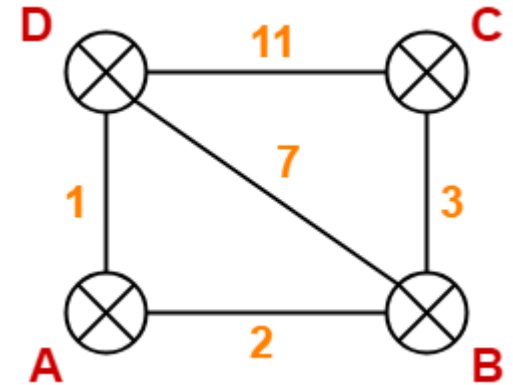
From D

1
7
11
0

New table, Router A

Destn	Dist	Nxt hop
A	0	A
B	2	B
C	5	B

- Cost of reaching router D from router A **via neighbor D** = Cost (A→D) + Cost (D→D) = **1 + 0**
- Cost of reaching router D from router A **via neighbor B** = Cost (A→B) + Cost (B→D) = **2 + 7**
- **min {1+0, 2+7} = 1 via D**



Old table, Router A

Destn	Dist	Nxt hop
A	0	A
B	2	B
C	∞	—
D	1	D

From B

A	2
B	0
C	3
D	7

From D

1
7
11
0

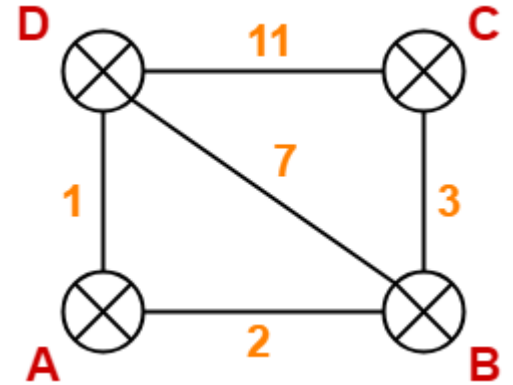
New table, Router A

Destn	Dist	Nxt hop
A	0	A
B	2	B
C	5	B
D	1	D

Router B

- Router B receives distance vectors from its neighbors A, C and D.
- Router B prepares a new routing table as-

	From A	From C	From D
A	0	∞	1
B	2	3	7
C	∞	0	11
D	1	11	0



Cost of reaching destination A from router B = $\min \{ 2+0, 3+\infty, 7+1 \} = 2$ via A.
Cost of reaching destination C from router B = $\min \{ 2+\infty, 3+0, 7+11 \} = 3$ via C.
Cost of reaching destination D from router B = $\min \{ 2+1, 3+11, 7+0 \} = 3$ via A

New table, Router B

Destn	Dist	Nxt hop
A	2	A
B	0	B
C	3	C
D	3	A

From B

2
0
3
7

From D

1
7
11
0

New table, Router C

Destn	Dist	Nxt hop
A	5	B
B	3	B
C	0	C
D	10	B

From A

0
2
∞
1

From B

2
0
3
7

From C

∞
3
0
11

New table, Router D

Destn	Dist	Nxt hop
A	1	A
B	3	A
C	10	B
D	0	D

- Step – 3

- Each router exchanges its distance vector obtained in Step-2 with its neighboring routers.
- After exchanging the distance vectors, each router prepares a new routing table

From B

2
0
3
3

From D

1
3
10
0

New table, Router A

Destn	Dist	Nxt hop
A	0	A
B	2	B
C	5	B
D	1	D

New table, Router B

Destn	Dist	Nxt hop
A	2	A
B	0	B
C	3	C
D	3	A

New table, Router C

Destn	Dist	Nxt hop
A	5	B
B	3	B
C	0	C
D	6	B

New table, Router D

Destn	Dist	Nxt hop
A	1	A
B	3	A
C	6	A
D	0	D

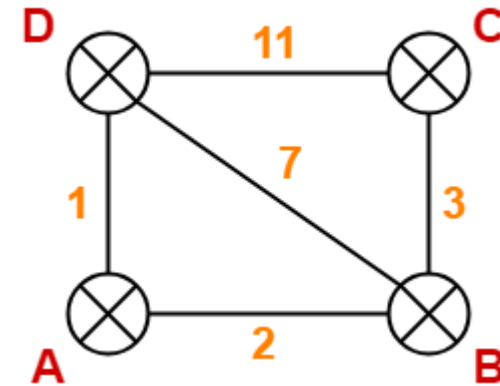
- Identifying Unused Links

- After routing tables converge (becomes stable),
 - Some of the links connecting the routers may never be used.
 - In the example, we can identify the unused links as,

New table, Router A

Destn	Dist	Nxt hop
A	0	A
B	2	B
C	5	B
D	1	D

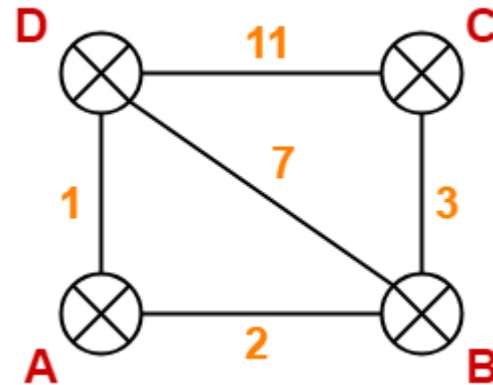
AB and AD are used.



New table, Router B

Destn	Dist	Nxt hop
A	2	A
B	0	B
C	3	C
D	3	A

BA and BC are used.



New table, Router C

Destn	Dist	Nxt hop
A	5	B
B	3	B
C	0	C
D	6	B

CB is used

New table, Router D

Destn	Dist	Nxt hop
A	1	A
B	3	A
C	6	A
D	0	D

DA is used.

Thus, edges BD and CD are never used.